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MUSA (MUMBAI UNIVERSITY STUDENTS ASSOCIATION)

VIVA QUESTION AND ANSWER OF BUSINESS INTELLIGENCE

T.E SEM-VI **SCHEME: R-19 C SCHEME** **FOR SUMMER SESSION 2026**

BRANCH: IT **BATCH: REGULAR**

MODULE 1: Star & Snowflake Schema (BI Lab)

Q.1) What is a Star Schema? How is it different from Snowflake Schema?
Ans: Star schema has a central fact table connected to denormalized dimension tables, while snowflake schema normalizes dimensions into multiple tables. Star is simpler; snowflake saves space but increases complexity.

Q.2) What is a fact table and dimension table?
Ans: Fact table stores measurable data (sales, revenue), while dimension tables store descriptive attributes (time, location). They together support analytical queries.

Q.3) Why is dimensional modeling used in BI systems?
Ans: Dimensional modeling improves query performance and simplifies data analysis. It is designed for OLAP operations and decision-making.

Q.4) What are advantages of Star Schema in BI?
Ans: It provides faster query performance, simple structure, and easy data understanding. It is widely used in data warehousing.

Q.5) When should Snowflake Schema be preferred?
Ans: Snowflake schema is preferred when data redundancy must be reduced and storage optimization is required, despite increased query complexity.

Q.6) What is OLAP and how does it relate to schema design?
Ans: OLAP (Online Analytical Processing) enables multidimensional analysis. Star and snowflake schemas are designed to support OLAP operations efficiently.

Q.7) Explain OLTP vs OLAP briefly.
Ans: OLTP handles transactional operations with frequent updates, while OLAP handles analytical queries for decision-making using historical data.

Q.8) What are dimensions in BI? Give examples.
Ans: Dimensions describe data context like time, product, and location. They help in analyzing facts from multiple perspectives.

Q.9) What is data warehouse? How is it related to BI?
Ans: A data warehouse stores integrated, historical data for analysis. BI uses this data to generate insights and support decisions.

Q.10) What is normalization vs denormalization in schema design?
Ans: Normalization reduces redundancy (used in snowflake), while denormalization improves query performance (used in star schema).

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MODULE 2: Data Exploration & Data Preprocessing

Q.1) What is data exploration? Why is it important before data mining?

Ans: Data exploration involves analyzing data to understand its structure, patterns, and anomalies. It helps identify issues like missing values and guides preprocessing steps.

Q.2) What are different types of attributes in data mining?

Ans: Attributes include nominal, ordinal, interval, and ratio types. They determine how data is processed and which algorithms can be applied.

Q.3) Explain statistical description of data with examples.

Ans: Statistical description includes mean, median, mode, variance, and standard deviation. These measures summarize data distribution and central tendency.

Q.4) What is data preprocessing? Explain its need.

Ans: Data preprocessing transforms raw data into clean and usable form. It is needed to handle noise, missing values, and inconsistencies for accurate results.

Q.5) What are data cleaning techniques?

Ans: Data cleaning includes handling missing values, removing noise, correcting inconsistencies, and smoothing data. It improves data quality.

Q.6) Explain data integration and data reduction techniques.

Ans: Data integration combines data from multiple sources, while data reduction reduces data size using techniques like sampling, aggregation, and feature selection.

Q.7) What is data transformation? Explain normalization.

Ans: Data transformation converts data into suitable format. Normalization scales data to a specific range, improving algorithm performance.

Q.8) What is discretization? Why is it used?

Ans: Discretization converts continuous data into discrete intervals. It simplifies data and improves interpretability for certain algorithms.

Q.9) Explain similarity and dissimilarity measures.

Ans: Similarity measures how alike data points are, while dissimilarity measures differences. They are used in clustering and classification.

Q.10) How is preprocessing performed in tools like WEKA or Python?

Ans: In WEKA, filters are used for cleaning and transformation, while in Python, libraries like Pandas and NumPy handle preprocessing tasks efficiently.

MODULE 3: Classification, Clustering & Frequent Pattern Mining

Q.1) What is classification? Explain with an example and algorithm.

Ans: Classification assigns data to predefined classes using labeled data. For example, spam detection using Decision Tree or Naïve Bayes.

Q.2) Differentiate between classification and clustering.

Ans: Classification is supervised (uses labeled data), while clustering is unsupervised (groups similar data without labels). Classification predicts classes; clustering discovers patterns.

Q.3) Explain Decision Tree and its advantages.

Ans: Decision Tree splits data based on attribute values to classify outcomes. It is easy to understand, interpretable, and handles both numerical and categorical data.

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Q.4) What is Naïve Bayes classifier? Why is it used?
Ans: Naïve Bayes is a probabilistic classifier based on Bayes theorem assuming feature independence. It is simple, fast, and effective for large datasets.

Q.5) What is clustering? Explain K-Means algorithm.
Ans: Clustering groups similar data points together. K-Means partitions data into K clusters by minimizing distance between points and cluster centroids.

Q.6) What are limitations of K-Means clustering?
Ans: K-Means requires predefined K, is sensitive to initial centroids, and struggles with non-spherical clusters and noise.

Q.7) What is frequent pattern mining? Explain Apriori algorithm.
Ans: Frequent pattern mining finds recurring patterns in data. Apriori generates frequent itemsets using candidate generation and pruning based on support.

Q.8) What is support and confidence in association rules?
Ans: Support measures frequency of itemset occurrence, while confidence measures the strength of rule implication. Both evaluate rule usefulness.

Q.9) How are these algorithms implemented in Python or WEKA?
Ans: In Python, libraries like Scikit-learn are used for classification and clustering, while WEKA provides built-in tools for applying and evaluating these algorithms.

Q.10) How do you evaluate performance of classification and clustering?
Ans: Classification is evaluated using accuracy, precision, recall, while clustering uses measures like silhouette score and intra-cluster distance.

MODULE 4: Implementation using BI Tools

Q.1) What is WEKA? How is it used in data mining?
Ans: WEKA is an open-source data mining tool used for classification, clustering, and association mining. It provides GUI-based implementation of algorithms without coding.

Q.2) Explain steps to perform classification in WEKA.
Ans: Load dataset → choose classifier (e.g., J48, Naïve Bayes) → select test option (training/test split) → run model → analyze output like accuracy and confusion matrix.

Q.3) How is clustering performed in WEKA?
Ans: Load dataset → select clustering algorithm (e.g., K-Means) → set parameters (number of clusters) → run → analyze cluster assignments and centroids.

Q.4) How do you perform association rule mining in WEKA?
Ans: Load dataset → select Apriori algorithm → set support/confidence thresholds → run → analyze generated association rules.

Q.5) What is confusion matrix? Why is it important?
Ans: Confusion matrix shows actual vs predicted classification results. It helps evaluate accuracy, precision, recall, and error rate.

Q.6) What is cross-validation in WEKA?
Ans: Cross-validation divides data into folds (e.g., 10-fold) to test model performance. It ensures reliable and unbiased evaluation.

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Q.7) What are advantages of using tools like WEKA over coding?
Ans: Tools are user-friendly, faster to implement, and require no programming. They are ideal for quick analysis and visualization.

Q.8) What are limitations of WEKA?
Ans: WEKA has limited scalability for large datasets and less flexibility compared to programming languages like Python.

Q.9) Compare WEKA with Python for data mining tasks.
Ans: WEKA is GUI-based and easy for beginners, while Python provides flexibility, scalability, and advanced customization using libraries.

Q.10) How do you interpret results generated by WEKA?
Ans: Results include accuracy, error rate, confusion matrix, and model summary. These metrics help evaluate model performance and decision-making.

MODULE 5: BI Tools Case Study

Q.1) What is Business Intelligence (BI)? How do BI tools help organizations?
Ans: BI refers to technologies and processes used to analyze data for decision-making. BI tools help visualize data, identify trends, and support strategic planning.

Q.2) What are common BI tools? Explain any one.
Ans: Common tools include Tableau, Pentaho, and QlikView. For example, Tableau is used for data visualization and creating interactive dashboards.

Q.3) What is Tableau? What are its key features?
Ans: Tableau is a data visualization tool that converts raw data into interactive dashboards. It supports drag-and-drop interface, real-time analysis, and multiple data sources.

Q.4) What is Pentaho? Where is it used?
Ans: Pentaho is a BI suite used for data integration, ETL, and reporting. It is used in enterprise-level data processing and analytics.

Q.5) What is QlikView and how is it different from Tableau?
Ans: QlikView is a BI tool focused on in-memory data processing and fast analytics, while Tableau focuses more on visualization and ease of use.

Q.6) What is data visualization? Why is it important in BI?
Ans: Data visualization represents data in graphical form like charts and dashboards. It helps in easy understanding and quick decision-making.

Q.7) What is a dashboard in BI tools?
Ans: A dashboard is a visual interface that displays key metrics and data insights in charts and graphs. It helps monitor performance and trends.

Q.8) Explain ETL process in BI tools.
Ans: ETL stands for Extract, Transform, Load. It collects data from sources, cleans/transforms it, and loads it into a data warehouse.

Q.9) How do BI tools support decision making?
Ans: BI tools analyze historical and real-time data to identify patterns and trends, helping managers make informed business decisions.

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Q.10) What are advantages and limitations of BI tools?
Ans: BI tools improve decision-making and visualization but may require high cost, training, and handling of large datasets.

MODULE 6: Business Intelligence Mini Project

Q.1) What is the objective of your BI project? How did you define the problem?
Ans: The objective is to analyze data and extract meaningful insights for decision-making. The problem is defined by identifying a real-world business need and selecting a suitable data mining task.

Q.2) How did you select the dataset and what are its features?
Ans: Dataset is selected from sources like Kaggle or UCI based on problem relevance. It contains attributes (features) and class labels used for analysis.

Q.3) What preprocessing steps did you apply on your dataset?
Ans: Preprocessing includes handling missing values, normalization, removing noise, and converting data into suitable format for algorithms.

Q.4) Which data mining technique did you use and why?
Ans: Techniques like classification, clustering, or association mining are used based on problem type. For example, classification is used for prediction tasks.

Q.5) Which algorithm did you implement and why?
Ans: Algorithms like Decision Tree, K-Means, or Apriori are selected based on efficiency and data nature. Choice depends on accuracy and problem requirements.

Q.6) How did you evaluate your model performance?
Ans: Performance is evaluated using metrics like accuracy, precision, recall, or clustering measures depending on the algorithm used.

Q.7) How did you visualize your results?
Ans: Results are visualized using graphs, charts, or dashboards in tools like Tableau or Python libraries to improve interpretation.

Q.8) What insights did you derive from your analysis?
Ans: Insights include patterns, trends, or relationships in data that help understand business behavior and improve decision-making.

Q.9) What BI decision did you suggest based on your project?
Ans: Based on analysis, decisions like improving sales strategy, customer targeting, or risk management are suggested for business growth.

Q.10) What challenges did you face during the project?
Ans: Challenges include data cleaning issues, selecting appropriate algorithms, and handling large datasets. These were resolved using preprocessing and proper tool selection.

*******ALL THE BEST*******

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